SIGNAL PROCESSING IV: Theories and Applications
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Speech Input Text Processing Using Multi-Stage Segmentation and Recovery Method

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This paper proposes a speech recognition method for a Japanese text processing system. In the speech input text processing, a recognition system intended for several ten-thousands words should be implemented. However it is still impossible to recognize all of them. In the proposed method, a speech waveform is recognized as a monosyllable string. Then the string is divided into a set of words. If there are some errors in the recognition, they are corrected during a word determination phase. When the monosyllable string is divided into some words, several possible words, i.e., word candidates, corresponding to certain monosyllable characters are taken out. The proposed method determines correct words from a set of word candidates using the statistics of appearance frequencies for the word candidates. In the experiment for a specific speaker, it is shown that the recognition rates 53.7% and 79.7% can be improved to 90.1% and 97.1% by the proposed method, respectively. In another experiment for an unspecified speaker, it is shown that we can recover recognition errors by using the same method, and that this method is effective for the Japanese speech text processing. In this paper, the method is called Multi-Stage Segmentation and Recovery Method.

1. INTRODUCTION.

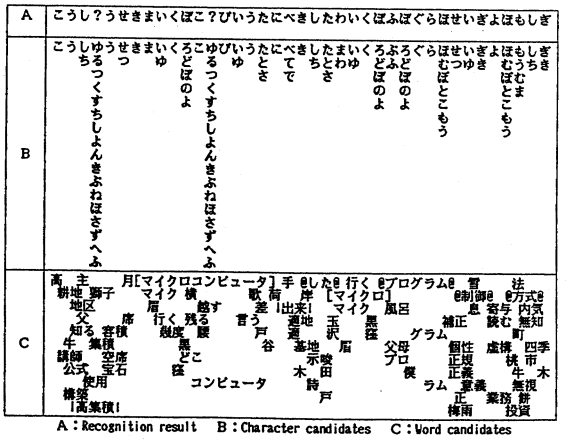
A speech input method for text processing is demanded in recent. In the speech input text processing, a recognition system for several ten-thousands words should be implemented. However any realized system for Japanese speech cannot recognize all of them [1]. In the proposed method, a speech waveform is recognized as a monosyllable string. Then the string is divided into a set of Japanese words. Each Japanese monosyllable corresponds to a Japanese phonetic character, i.e., "Kana" in Japanese. Therefore we can treat the recognized monosyllable strings as the "Kana" strings. The process for dividing monosyllable strings into words is similar to that for dividing non-segmented Kana-written strings into words. The methods for non-segmented "Kana"-written strings, i.e., for "Beta-gaki" sentences, have been already proposed [2]. Therefore we can consider the proposed process as the extension of the method for "Beta-gaki" sentences. But since there exist ambiguities or errors in the recognized monosyllable string, a larger number of word candidates will appear than the word candidates in the process of "Beta-gaki" sentences. The proposed method determines a set of correct words using the statistics values which are calculated from the appearance frequencies for word candidates. In this paper, the method is called Multi-Stage Segmentation and Recovery Method.

2. OUTLINES OF MULTI-STAGE SEGMENTATION AND RECOVERY METHOD.

From a recognized monosyllable string, the proposed method first determines a part which is associated with a specific word in the string, and then recovers errors except the determined part step by step.

Fig.1 shows an example for this process. In this method, a recognition result, i.e., "Beta-gaki" string, is first given (Fig.1 (1)-A where "?" denotes an unrecognized "Kana" character). Then, as shown in Fig.1 (1)-B, all character candidates are retrieved for a "Beta-gaki" string from a given dictionary owing to the following reason. The input monosyllable does not always correspond to a real speech because of some mistakes in the recognition. Thus, generally speaking, there are some character candidates for a recognized monosyllable. The correspondence between the input monosyllable and its recognition result have been already examined, and then it is stored in a dictionary. The dictionary which stores the information is called a recognition dictionary. All character candidates are taken out for the recognized

(1) Example for word candidates.



(2) Change of a monosyllable string.

Step	Result	D
A	こうし?うせきまいくぼこ?びいうたにべきしたわいくぼふぼぐらほせいぎよほもしぎ	92
1	こうし?うせきまいくぼこ?びいうたにべきし@したわいくぼ@プログラム@@制御@@方式@	52
2	こうし?うせき[マイクロコンピュータ]にべきし@した@[マイクロ]@プログラム@@制御@@方式@	30
3	|高集積|[マイクロコンピュータ]に|出来|@した@[マイクロ]@プログラム@@制御@@方式@	9
4	|高集積|[マイクロコンピュータ](に)|出来|@した@[マイクロ]@プログラム@@制御@@方式@	8

1: The first reank key-words @ @ 2: The scond rank key-words [ ] 3: Ordinary words | |
4: One character particles ( ) D: The total number of word candidates

Fig.1 Example of recovery process.

monosyllable from the recognition dictionary as mentioned above. For example, when "た" is recognized from speech, the character candidates are given as "た", "と" and "さ" in Fig.1 (1)-B. After that, many word candidates are taken out by combining character candidates (Fig.1 (1)-C). A set of correct words are chosen among them. During the process, this method determines the part which can be fixed by a specific word in word candidates first, and determines correct words step by step. The specific word has been already registered in dictionaries. The words which are first determined are called "key-words". In Fig.1, "した", "プログラム", "制御", "方式", "マイクロコンピュータ", and "マイクロ" have been already stored as key-words in key-word dictionaries. Thus these words are fixed as correct words for the "Beta-gaki" string (Fig.1 (2)-1 and -2). The key-words have been extracted from a series of articles in a specific field and stored in the key-word dictionaries [2]. A "Beta-gaki" sentence is first divided into several blocks by key-words, then the division is continued step by step using a low-rank word set (Fig.1 (2)-3 and -4), and finally the "Beta-gaki" sentence is completely divided into a unique word string.

The number of candidate words determined in the low-rank sets rapidly decreases since the key-words are first determined as correct words. Therefore the method is considerably effective for the process of a speech input "Beta-gaki" sentence which contains a large amount of character ambiguities.

3. EXPERIMENTAL SYSTEM.

3.1 Construction of system.

Fig.2 shows the construction of an experimental system. This system consists of two subsystems which are a speech recognition part and a recovery process part.

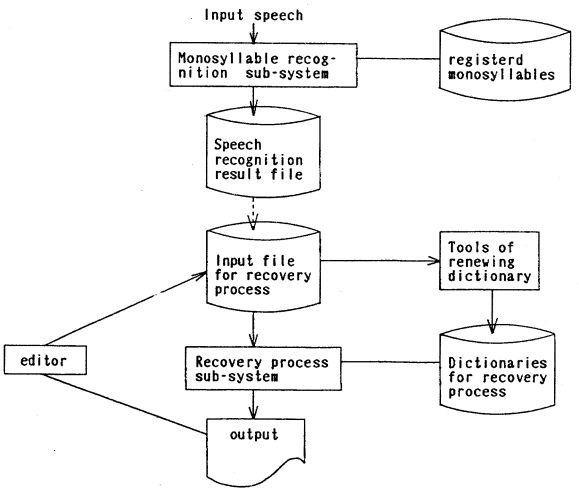


Fig.2 Construction of the system.

3.2 Recovery algorithm in the recovery process part.

All words are classified into some ranks according to their statistics. The recovery algorithm uses these ranks. Fig.3 shows the recovery process. The details of each step is given as follows:

(1) Division and recovery by the first rank and second rank key-words.

The key-words are defined as the words which the system can determine as correct words unconditionally when their "Kana" characters correspond to that of "Beta-gaki" sentences. All key-words have been already extracted from several articles in a specific field and stored in the key-word dictionaries. The key-words are classified into two kinds of words which are the first rank key-words and the second rank key-words. The first rank key-words are used independently of any article. The different second rank key-words are served for each article.

(2) Key-words extension.

The system searches a new word which correspond to a key-word and some "Kana" characters near to the key-word. If it succeeds, the new word is determined instead of the key-word and the "Kana" characters. Otherwise, step 2 is skipped. For example, if a key-word is given as "処理" and the "Beta-gaki" string around it is given as "・・・・みしょり・・・・" where "処理" corresponds to "しょり", then this step determines "未処理(みしょり)" instead of them.

(3) Division and recovery by "Katakana" words.

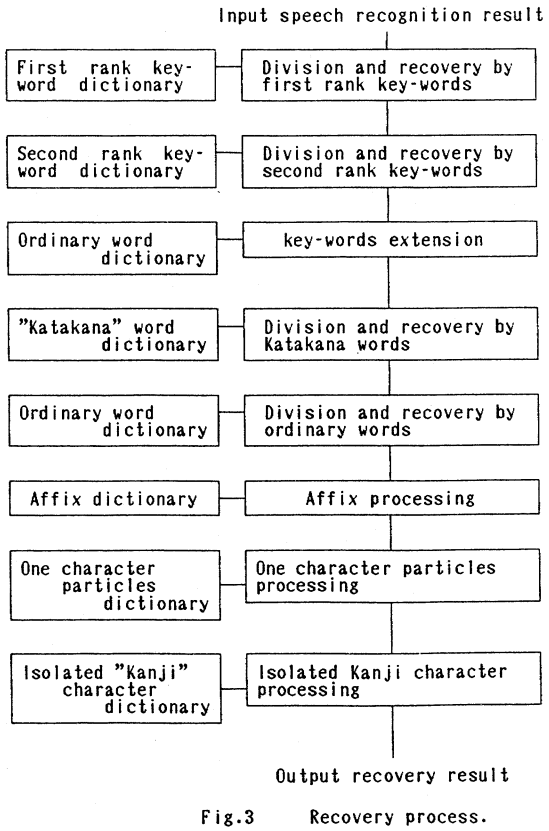


Fig.3 Recovery process.

Usually, foreign words or onomatopoeic words are expressed with "Katakana" characters and are called "Katakana" words. Foreign words and onomatopoeic words are correct words with the highest possibility except key-words since the characteristics in the expression of foreign and onomatopoeic words are different from the others. Therefore, the system divides "Beta-gaki" sentences by the "Katakana" words after the division by key-words.

(4) Division and recovery by ordinary words.

In this step, there are many word candidates of which the "Kana" character correspond to the part of "Beta-gaki" sentences since the certainty of words fairly decreases. Therefore, for each word, we define a triplet which includes the word, a "Kana" character just front it and a "Kana" character just behind it. The ordinary word dictionary memorizes the appearance frequency of each triplet. This step first searches the triplet which exists both in the "Beta-gaki" sentence and in the ordinary word dictionary, and then determines the triplet with the highest appearance frequency.

(5) Affix processing.

The system determines affix candidates as correct affixes if their "Kana" characters correspond to an unfixed part front or behind "Kanji" words, "Katakana" words or figures in the "Beta-gaki" sentences.

(6) One character particles processing.

In this paper, "Zyosi" in Japanese is called particle. Among all particle candidates, the system determines one character particle as a correct particle if both of parts front and behind it are fixed.

(7) Isolated "Kanji" character processing.

In the last step, if the unfixed part of a "Beta-gaki" sentence corresponds to the phonetic characters of a isolated "Kanji" character candidate, the system determines it as correct isolated "Kanji" character.

4. EXPERIMENT FOR PERFORMANCE EVALUATION.

4.1 Experiment of recover for specific speaker speech.

Experimental procedure is given as follows:

(1-1) A specific speaker utters 68 monosyllables to the monosyllable speech recognition system [3] as training. The system recognizes the 68 monosyllables and registers their results in the voice dictionary.

Table 1 Data for the recovery experiment.

No	Total number of monosyllables
1	474
2	395
Total	869

Source: Chapter one in "A Methodology of Performance Management with an Experimental Result", IPSJ Trans. vol.23, no.6, pp.591-598, 1982.

(1-2) The same speaker utters the data No.1 or No.2 of table 1 and the system memorizes the recognized results.

(1-3) We produce the five kinds of data from those results by correcting their errors randomly in order to increase the simulated recognition rates to 70%, 75%, 80%, 85% and 90%.

(1-4) The recovery experiment is performed by each set of using simulation data and the recognized result.

The initial conditions of the dictionaries are given as follows:

(2-1) The data for the dictionaries except the recognition dictionary.
The data used in this experiment are 14 papers related to the information processing. The total number is 120,741 characters. These dictionaries are renewed on each paper by using the Kana-Kanji translation system with Multi-Stage Segmentation Method for "Beta-gaki" sentences [2].

(2-2) The data for the recognition dictionary.

The dictionary is made from each data given in table 1. The correspondences between real input monosyllables and their recognition results are stored.

Table 2 shows the results of the recovery experiment. If we define
A: the total number of correct monosyllables
B: the total number of erroneous monosyllables
C: the total number of unfixed monosyllables
D: all number of uttered monosyllables
where an unfixed monosyllable is defined as a monosyllable which is not determined through the recovery process, then a correct rate, an erroneous rate and an unfixed rate are given as A/D, B/D and C/D, respectively.

Table 2 Results of the recovery experiment for a specific speaker speech.

Given data		Correct		Erroneous		Unfixed	
		Frequency	Rate %	Frequency	Rate %	Frequency	Rate %
Recognition data	A	467	53.7	272	31.3	130	15.0
	B	693	79.7	172	19.8	4	0.5
Simulation data	A	618	71.1	165	19.0	86	10.0
	B	766	88.1	99	11.4	4	0.5
	A	659	75.8	138	15.9	72	8.3
	B	781	89.9	86	9.9	2	0.2
	A	699	80.4	106	12.2	64	7.4
	B	793	91.3	75	8.6	1	0.1
	A	739	85.0	80	9.2	50	5.8
	B	816	93.9	52	6.0	1	0.1
	A	783	90.1	56	6.4	30	3.5
	B	844	97.1	24	2.8	1	0.1

A : The frequency and rate before the recovery experiment
B : The frequency and rate after the recovery experiment

In the recovery for a specific speaker, it is shown that the recognition rate 53.7% and 90.1% in the speech recognition stage can be improved to 79.7% and 97.1% in the word determination stage by Multi-Stage Segmentation and Recovery Method, respectively.

The speech recognition system for a monosyllable of a specific speaker with the recognition rate of 90% can be realized generally [1]. The recognition rate about 90% can be improved to more than 95% by this proposed method. The mimimum requirement for the translation rate of Japanese word processing using a keyboard is said to be more than 95% [4]. Consequently, the proposed method using speech input fills the above-mentioned request.

4.2 Experiment of recover for unspecified speaker speech.

The ability of recovery for the errors of an unspecified speaker is examined by using this method. We think that this method can recover the errors of an unspecified speaker speech since the tendency of errors in the recognition system for an unspecified speaker is similar to that for a specific speaker.

The registered procedure is the same as 4.1 (1-1). An unspecified speaker utters the data of table 1.

The data used the experiment is the same as the recovery experiment of a specific speaker speech.

Table 3 shows the results of the recovery experiment of speech spoken by an unspecified speaker. The definitions of a correct rate, an erroneous rate and an unfixed rate are the same as those in recovery experiment of a specific speaker.

According to table 3, the recognition rate of an unspecified speaker speech can be improve from 28.5% to 63.6%. Since the recognition rate of an unspecified speaker speech before the recovery is 28.5%, it is shown that this method can be applied to recover an unspecified speaker speech.

Table 3 Results of the recovery experiment for an unspecified speaker speech.

	A		B	
	Frequency	Rate %	Frequency	Rate %
Correct	248	28.5	553	63.6
Erroneous	451	51.9	309	35.6
Unfixed	170	19.6	7	0.8

A : The frequency and rate before the recovery experiment
B : The frequency and rate after the recovery experiment

5. CONCLUSION.

In the speech recognition for Japanese text processing, there are extremely a large number of word candidates since there are ambiguities or errors in recognized speech. Therefore, it is necessary to select correct words among them. We have proposed Multi-Stage Segmentation and Recovery Method for solving the problem. In this method, the system recognizes speech at each monosyllable and takes out all character candidates from the recognition results. Then the system determines a set of correct words from word candidates according to their possibility order. We have developed an experimental system by this method and carried out experiments for the performance evaluation. As experimental results, it is shown that the recognition rate 53.7% and 90.1% can be improved to 79.7% and 97.1%, respectively. In addition we have performed the other experiment in which some errors of an unspecified speaker are recovered. It is shown that this proposed method can be applied to the speech input Japanese text processing for an unspecified speaker.

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